

Amelia Nguyen

248-343-0064 | amelianguyen.ds@gmail.com | [linkedin.com/in/ameliang12/](https://www.linkedin.com/in/ameliang12/) | amelia-ng.github.io/portfolio/

EDUCATION

University of Michigan - Dearborn

Master's in Data Science, GPA: 4.0

Dearborn, MI

Aug 2025 – May 2027

EXPERIENCE

University of Michigan

Graduate Research and Teaching Assistant

Dearborn, MI

Jun 2026 – Present

- **Research Assistant:** Leading a 6-member team researching an AI-driven indoor localization system using visible light communication; co-authored 3 additional conference publications on AI wireless communication and sensing systems.
- **AI Research:** Developed computer vision models for object detection from rolling-shutter LED signal data; leveraged Generative AI (LLMs, RAGs) to fine-tune, detect model anomalies, and develop solutions, achieving 85%+ accuracy.
- **Teaching Assistant:** Graded assignments and exams and provided feedback for 30 students across CIS 400 Programming Languages and CIS 541 Immersive Computing and Digital Twins.

Gearlaunch

Data Analyst

Salt Lake City, UT

Nov 2023 – Jul 2025

- **Machine Learning:** Developed and deployed Regression, Random Forest, and XGBoost models for sales and business forecasting, contributing to a 20% increase in annual sales KPI. (Python, PostgreSQL, GCP)
- **Statistical Analytics:** Applied advanced data science, including statistical and time-series analysis, and Tableau to identify trends and forecast performance, delivering data-driven insights across 5 business functions.
- **Data Engineering:** Engineered scalable data pipelines and automated workflows using GCP (BigQuery, Cloud Run, Vertex AI), supporting large-scale transaction data from 35,000+ storefronts.
- **BI Analytics:** Owned end-to-end analytics as the sole analyst, delivering 10+ monthly executive reports and data-driven recommendations to C-level management, contributing to a 50% increase in marketing conversion rates.

VCCorp Corporation

Data Analyst - Team Lead

Hanoi

May 2023 – Nov 2023

- **Recommendation Systems:** Led a team of 4 to deploy and evaluate 25+ ML recommendation systems using PySpark, Python, SQL, and Airflow across 7 media platforms, driving 10–30% monthly core KPI growth.
- **Experimentation:** Built an automated A/B testing evaluation framework for 7 web platforms, enabling scalable experimentation and later productized as a client-facing analytics solution.
- **Data Engineering:** Optimized ETL pipelines with PySpark, HDFS, Airflow, and SQL to process and transform large-scale user and content data; used Git for collaborative development and version control.

Data Analyst

Mar 2021 – May 2023

- **Product Analytics:** Designed 20+ product KPIs to analyze user behavior to optimize engagement strategies across 7 digital media platforms, contributing to 10% monthly product growth.
- **Anomaly Detection:** Applied statistical and ML techniques, including multivariate outlier detection, clustering, and isolation forest, to identify abnormalities in notification patterns and user behavior.

TECHNICAL SKILLS

AI & ML: Regression, Decision Trees, Random Forests, XGBoost, LightGBM, SVM, KNN, Clustering, PCA, Deep Learning, CNN, Time Series, Computer Vision, Transformer, NLP, Reinforcement Learning, LLMs, Generative AI, Agentic AI, RAG

Language & Frameworks: Python (Pandas, Scikit-learn, NumPy, PyTorch, TensorFlow, Matplotlib, Seaborn), SQL, PySpark

Cloud & MLOps: Git, Docker, Google Cloud Platform, AWS, Airflow, CI/CD

Analytics & FrontEnd: Tableau, Power BI, Google Analytics, SAP Analytics Cloud, JavaScript, HTML

PROJECTS

Vehicle Defect Detection | *PyTorch, FastAPI, Computer Vision, Docker, AWS*

Jul 2026

Traffic Obstacle Detection | *PyTorch, Computer Vision, Hugging Face*

Jun 2026

Fuel Consumption Prediction for Real-World OBD-II Data | *Python, XGBoost, LightGBM, LSTM*

May 2026

Vehicle Crash Frequency and Severity Prediction | *Python, Regression, Classification, PCA, LDA*

Dec 2025

PUBLICATIONS

D. Acikbas, A. Jayeb, Y. F. Owusu, **T. A. Nguyen**, C. Friese, K. Parmenter, H. Fynn, N. Mao, Z. Zhou, X. Zhang, “Multi-Modal Contactless Sensing for Lithium Battery SOC Estimation”, *Under Review - IEEE WCNC*, 2026.

H. Fynn, **T. A. Nguyen**, C. Nwobu, A. Sarvadey, T. Li, J. Wang, K. Sun, X. Zhang, “TAW-Ring: Low-Cost Passive Wearable Optical Ring for Unclonable Revocable Mobile Authentication”, *Under Review - ACM MobiCom*, 2027.

D. Acikbas, C. Nwobu, H. E. Fynn, **T. A. Nguyen**, K. Kattan, B. Maxim, X. Zhang, “Rolling-Shutter Wrist Micro-Movement Sensing for Privacy-Preserving Emotion Recognition”, *ACM MobiCom Demo*, 2026.